

Radiative Transfer: Exchange between Diffuse Surface

Master

Cyril Caliot (CNRS)

LMAP, Laboratoire de Mathématiques et de leurs Applications de Pau,
1 Allée du Parc Montauray, 64600 Anglet

cyril.caliot@univ-pau.fr



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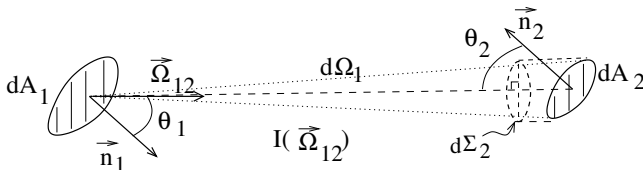
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Exchanged Radiative Flux between 2 Elementary Surfaces

The elementary flux $d^2\Phi$ emitted by dA_1 inside a solid angle $d\Omega_1$ around direction $\vec{\Omega}_{12}$ is expressed by:

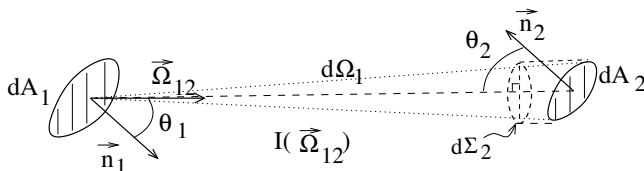
$$\begin{aligned} d^2\Phi_{1\rightarrow 2} &= I(\vec{\Omega}_{12}) d\Sigma_1 d\Omega_1 \\ &= I(\vec{\Omega}_{12}) |\vec{n}_1 \cdot \vec{\Omega}_{12}| dA_1 d\Omega_1 \\ &= I(\vec{\Omega}_{12}) |\vec{n}_1 \cdot \vec{\Omega}_{12}| dA_1 \frac{d\Sigma_2}{R_{12}^2} \end{aligned}$$

$$\begin{aligned} d^2\Phi_{1\rightarrow 2} &= I(\vec{\Omega}_{12}) |\vec{n}_1 \cdot \vec{\Omega}_{12}| dA_1 \frac{|\vec{n}_2 \cdot \vec{\Omega}_{12}| dA_2}{R_{12}^2} \\ &= I(\vec{\Omega}_{12}) \frac{\cos(\theta_1) \cos(\theta_2) dA_1 dA_2}{R_{12}^2} \end{aligned}$$



$\vec{n}_1$, $\vec{n}_2$ and $\vec{\Omega}_{12}$ are unit vectors; $R_{12} = \|\vec{x}_2 - \vec{x}_1\|$.

Intercepted Fluxes



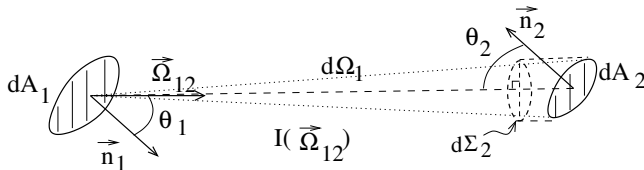
Leaving Flux from A_1

$$\Phi_{A_1}^l = \int_0^\infty \int_{2\pi} \int_{A_1} I_1^l(\lambda, \vec{\Omega}) |\vec{\Omega} \cdot \vec{n}_1| dA_1 d\Omega d\lambda$$

Leaving Flux from A_1 and intercepted by A_2

$$\Phi_{A_1 \rightarrow A_2} = \int_0^\infty \int_{A_2} \int_{A_1} I_1^l(\lambda, \vec{\Omega}) |\vec{\Omega} \cdot \vec{n}_1| dA_1 \frac{|\vec{\Omega} \cdot \vec{n}_2|}{R^2} dA_2 d\lambda$$

Exchanged NET Fluxes between two Surfaces



Exchanged Net Flux between A_1 and A_2 : $\Phi_{12} = \Phi_{A_1 \rightarrow A_2} - \Phi_{A_2 \rightarrow A_1}$

$$\Phi_{12} = \int_0^\infty \int_{A_2} \int_{A_1} \left(I_1^l(\lambda, \vec{\Omega}_{12}) - I_2^l(\lambda, -\vec{\Omega}_{12}) \right) |\vec{\Omega} \cdot \vec{n}_1| dA_1 \frac{|\vec{\Omega} \cdot \vec{n}_2|}{R^2} dA_2 d\lambda$$

with

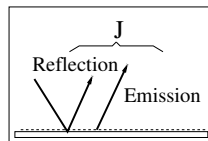
$$I_1^l(\lambda, \vec{\Omega}_{12}) = \varepsilon'(\lambda, \vec{\Omega}_{12}) I_b(T_1, \lambda) + \int_{2\pi} \rho''(\lambda, \vec{\Omega}_{12}, \vec{\Omega}) I(\lambda, \vec{\Omega}) |\vec{\Omega} \cdot \vec{n}_1| d\Omega$$

Net Flux Exchanged between 2 surfaces OPAQUE, GRAY and DIFFUSE

If the surface is **OPAQUE, GRAY and DIFFUSE** ($\rho'' = \rho/\pi$) then

$$\begin{aligned} I_1^l(\vec{\Omega}_{12}) &= \epsilon I_b(T_1) + \frac{\rho}{\pi} \int_{2\pi} I(\vec{\Omega}) |\vec{\Omega} \cdot \vec{n}_1| d\Omega \\ &= \frac{1}{\pi} (\epsilon E_b(T_1) + \rho H_1) = \frac{1}{\pi} J_1 \end{aligned}$$

with J the **radiosity**, $J = E + \rho H$
($W.m^{-2}$)



$$\Phi_{12} = \int_{A_2} \int_{A_1} \left(\frac{J_1}{\pi} - \frac{J_2}{\pi} \right) \frac{|\vec{\Omega} \cdot \vec{n}_1| |\vec{\Omega} \cdot \vec{n}_2|}{R^2} dA_1 dA_2$$

Net Flux Exchanged between 2 surfaces: Radiosity

$$\begin{aligned}
 \Phi &= \Phi_{\text{emitted}} - \Phi_{\text{absorbed}} = A \left[E - \alpha H \right] \\
 &= A \left[E - (1 - \rho) H \right] = A \left[(E + \rho H) - H \right] \\
 &= A \left[J - H \right] = (\Phi_{\text{emitted}} + \Phi_{\text{reflected}}) - \Phi_{\text{incident}}
 \end{aligned}$$

- if $\Phi > 0$, A loses energy by radiation,
- if $\Phi = 0$, the losses by emission from A equals the absorbed fluxes coming from the surrounding surfaces.
- si $\Phi < 0$, A gains energy by radiation.

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View Factors for Surfaces OP., GR. and DIFF.

Assuming each radiosity is constant over its surface (!):

$$\Phi_{12} = (J_1 - J_2) \int_{A_2} \int_{A_1} \frac{|\vec{\Omega} \cdot \vec{n}_1| |\vec{\Omega} \cdot \vec{n}_2|}{\pi R^2} dA_1 dA_2$$

Considering the net flux is proportional to A_1 , the view factor F_{12} is introduced:

$$\begin{aligned} \Phi_{12} &= (J_1 - J_2) A_1 F_{12} \\ F_{12} &= \frac{1}{A_1} \int_{A_2} \int_{A_1} \frac{|\vec{\Omega} \cdot \vec{n}_1| |\vec{\Omega} \cdot \vec{n}_2|}{\pi R^2} dA_1 dA_2 \end{aligned}$$

The view factor F_{12} is the fraction of radiative flux (emission+reflection) leaving A_1 and intercepted by A_2

Reciprocity Relation

$$dF_{dA_i \rightarrow dA_j} = \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}_j|}{\pi R^2} dA_j$$

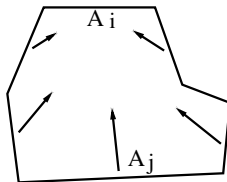
$$F_{dA_i \rightarrow A_j} = \int_{A_j} \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}_j|}{\pi R^2} dA_j$$

$$F_{A_i \rightarrow A_j} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}_j|}{\pi R^2} dA_j dA_i$$

$$F_{A_j \rightarrow A_i} = \frac{1}{A_j} \int_{A_j} \int_{A_i} \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}_j|}{\pi R^2} dA_i dA_j$$

$$A_i F_{i \rightarrow j} = \int_{A_i} \int_{A_j} \frac{\cos(\theta_i) \cos(\theta_j)}{\pi R^2} dA_i dA_j = A_j F_{j \rightarrow i}$$

Irradiation and View Factors



The radiation incident on surface A_i in a **cavity with opaque, diffuse and gray surfaces**:

$$\begin{aligned}
 A_i H_i &= \int_{A_i} \int_{2\pi \vec{n}_i \cdot \vec{\Omega} < 0} I(\vec{\Omega}) |\vec{\Omega} \cdot \vec{n}_i| d\Omega dA_i = \int_{A_i} \int_A J(\vec{\Omega}) \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}|}{\pi R^2} dA dA_i \\
 &= \int_{A_i} \sum_{j=1}^N J_j \int_{A_j} \frac{|\vec{\Omega} \cdot \vec{n}_i| |\vec{\Omega} \cdot \vec{n}_j|}{\pi R^2} dA_j dA_i \\
 &= \sum_{j=1}^N J_j A_i F_{i \rightarrow j} = \sum_{j=1}^N J_j A_j F_{j \rightarrow i}
 \end{aligned}$$

Relation in a Closed Cavity with Black Walls

The cavity is **closed**, the emitted flux in the hemisphere is entirely intercepted by the black surfaces:

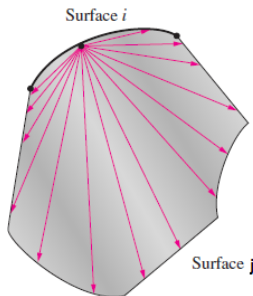
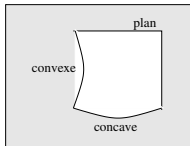
$$\Phi_i^e = \sum_{j=1}^N A_i E_{b,i} F_{i \rightarrow j} = A_i E_{b,i} \quad \Rightarrow \quad \sum_{j=1}^N F_{i \rightarrow j} = 1$$

- convex surfaces

$$F_{i \rightarrow i} = 0$$

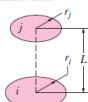
- concave surfaces

$$F_{i \rightarrow i} \neq 0$$



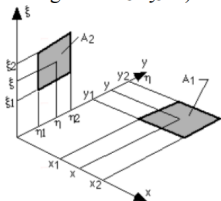
Analytical Formula for View Factors

View factor expressions for some common geometries of finite size (3D)

Geometry	Relation
Aligned parallel rectangles 	$\bar{X} = X/L \text{ and } \bar{Y} = Y/L$ $F_{i \rightarrow j} = \frac{2}{\pi \bar{X} \bar{Y}} \left\{ \ln \left[\frac{(1 + \bar{X}^2)(1 + \bar{Y}^2)}{1 + \bar{X}^2 + \bar{Y}^2} \right]^{1/2} \right. \\ + \bar{X}(1 + \bar{Y}^2)^{1/2} \tan^{-1} \frac{\bar{X}}{(1 + \bar{Y}^2)^{1/2}} \\ + \bar{Y}(1 + \bar{X}^2)^{1/2} \tan^{-1} \frac{\bar{Y}}{(1 + \bar{X}^2)^{1/2}} \\ \left. - \bar{X} \tan^{-1} \bar{X} - \bar{Y} \tan^{-1} \bar{Y} \right\}$
Coaxial parallel disks 	$R_i = r_i/L \text{ and } R_j = r_j/L$ $S = 1 + \frac{1 + R_i^2}{R_j^2}$ $F_{i \rightarrow j} = \frac{1}{2} \left\{ S - \left[S^2 - 4 \left(\frac{r_j}{r_i} \right)^2 \right]^{1/2} \right\}$
Perpendicular rectangles with a common edge 	$H = Z/X \text{ and } W = Y/X$ $F_{i \rightarrow j} = \frac{1}{\pi W} \left(W \tan^{-1} \frac{1}{W} + H \tan^{-1} \frac{1}{H} \right. \\ - (H^2 + W^2)^{1/2} \tan^{-1} \frac{1}{(H^2 + W^2)^{1/2}} \\ + \frac{1}{4} \ln \left\{ \frac{(1 + W^2)(1 + H^2)}{1 + W^2 + H^2} \right\} \\ \times \left[\frac{W^2(1 + W^2 + H^2)}{(1 + W^2)(W^2 + H^2)} \right]^{W^2} \\ \times \left[\frac{H^2(1 + H^2 + W^2)}{(1 + H^2)(H^2 + W^2)} \right]^{H^2} \Bigg\}$

Analytical Formula for View Factors

From rectangle A_1 at 90° to rectangle A_2 (mind it is singular if $x_1 = \xi_1 = 0$).



$$F_{12} = \frac{1}{2\pi A_1} \sum_{l=1}^2 \sum_{k=1}^2 \sum_{j=1}^2 \sum_{i=1}^2 \left[(-1)^{(i+j+k+l)} B(x_i, y_j, \eta_k, \xi_l) \right]$$

$$\text{with } A_1 = (x_2 - x_1)(y_2 - y_1)$$

$$B = (y - \eta) C \arctan(D) - \frac{C^2}{4} (1 - D^2) \ln \left[C^2 (1 + D^2) \right]$$

$$C = \sqrt{x^2 + \xi^2}, \quad D = \frac{(y - \eta)}{C}$$

(e.g. for squares touching:

$$x_1 = \xi_1 = 10^{-6}, x_2 = \xi_2 = 1, y_1 = \eta_1 = 0, y_2 = \eta_2 = 1, F_{12} = 0.20004)$$

Catalogue : <http://www.engr.uky.edu/rtl/Catalog/>

View Factor Charts

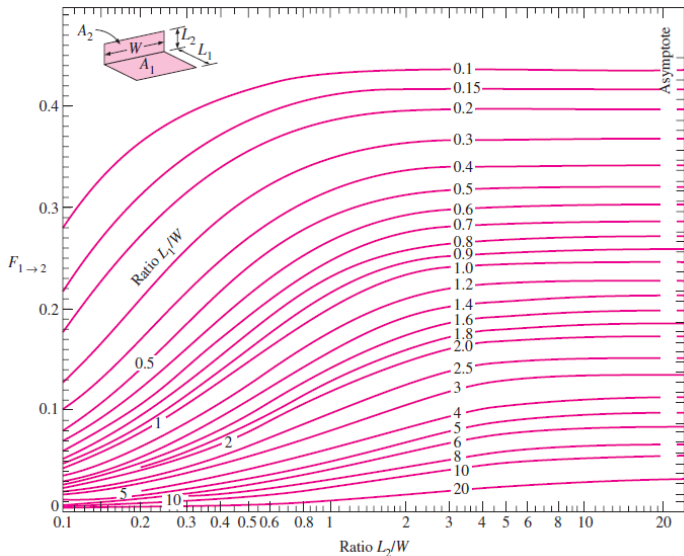


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Exchanged Net Flux between BLACK Surfaces

The net flux exchanged by A_i with the cavity is :

$$\Phi_i = A_i \left(E_b(T_i) - H_i \right)$$

The incident irradiation of surface A_i comes from all the surfaces A_j that emit towards A_i .

$$A_i H_i = \sum_{j=1}^N A_j F_{ji} E_b(T_j)$$

Using the reciprocity relation between the view factors:

$$H_i = \frac{1}{A_i} \sum_{j=1}^N A_i F_{ij} E_b(T_j) = \sum_{j=1}^N F_{ij} E_b(T_j)$$

$$\Phi_i = A_i \left(E_b(T_i) - \sum_{j=1}^N F_{ij} E_b(T_j) \right)$$

Closed Cavity with N BLACK Surfaces

Net exchanged flux by A_i :

$$\Phi_i = A_i \left(E_{b,i} - \sum_{j=1}^N F_{ij} E_{b,j} \right)$$

Noting that $E_{b,i} = E_{b,i} \sum_{j=1}^N F_{ij} = \sum_{j=1}^N F_{ij} E_{b,i}$

$$\begin{aligned} \Phi_i &= A_i \left(\sum_{j=1}^N F_{ij} E_{b,i} - \sum_{j=1}^N F_{ij} E_{b,j} \right) \\ &= A_i \sum_{j=1}^N F_{ij} \left(E_{b,i} - E_{b,j} \right) \\ &= A_i \sigma \sum_{j=1}^N F_{ij} \left(T_i^4 - T_j^4 \right) \end{aligned}$$

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Exch. Net Flux for OP., GRAY and DIFF. Surfaces

The Net Flux Φ_i : (radiative thermal balance at the surface A_i)

$$\Phi_i = A_i [J_i - H_i] = A_i J_i - A_i H_i$$

$$A_i J_i = A_i (E_i + \rho_i H_i) = A_i \epsilon_i E_{b,i} + (1 - \epsilon_i) A_i H_i$$

$$A_i H_i = A_i J_i - \Phi_i$$

We want to simplify the irradiation in the expression of the net flux.

$$\Phi_i = A_i \epsilon_i E_{b,i} - \epsilon_i [A_i J_i - \Phi_i]$$

$$\Phi_i (1 - \epsilon_i) = A_i [\epsilon_i E_{b,i} - \epsilon_i J_i]$$

$$\Phi_i = A_i \frac{\epsilon_i}{1 - \epsilon_i} [E_{b,i} - J_i] \quad \text{with } (\epsilon_i \neq 1)$$

Surfaces with Imposed Temperatures

$E_{b,i}$ is known, we will use:

$$\frac{\Phi_i}{A_i} = J_i - H_i = \frac{\epsilon_i}{1 - \epsilon_i} \left[E_{b,i} - J_i \right]$$

It will be possible to express all with the radiosity:

$$J_i - \sum_{j=1}^N F_{ij} J_j = \frac{\epsilon_i}{1 - \epsilon_i} \left[E_{b,i} - J_i \right]$$

Then, a linear system with N equations and N unknowns will be solved.

Matrix form notation **AJ=B**:

$$\begin{aligned} A_{ij} &= \delta_{ij} - (1 - \epsilon_i) F_{ij} \\ B_i &= \epsilon_i E_{b,i} = \epsilon_i \sigma T_i^4 \end{aligned}$$

Surfaces with Imposed Fluxes

The net flux Φ_i is known, we will use:

$$\frac{\Phi_i}{A_i} = J_i - H_i = J_i - \sum_{j=1}^N F_{ij} J_j$$

Then, a linear system with N equations and N unknowns will be solved.
Matrix form notation $\mathbf{A}\mathbf{J}=\mathbf{B}$:

$$\begin{aligned} A_{ij} &= \delta_{ij} - F_{ij} \\ B_i &= \frac{\Phi_i}{A_i} \end{aligned}$$

Temperature and Flux Computation

Once the radiosities are computed, the net fluxes or the temperatures may be found by:

$$\Phi_i = \sum_{j=1}^N \left[\delta_{ij} - F_{ij} \right] J_j$$

$$T_i = \left[\frac{1}{\sigma \epsilon_i} \left(\sum_{j=1}^N \left[\delta_{ij} - (1 - \epsilon_i) F_{ij} \right] J_j \right) \right]^{\frac{1}{4}}$$

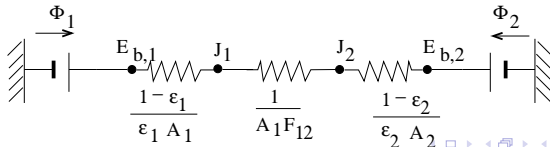
Electrical Analogy: Gray Surfaces

The net flux may be written:

$$\frac{\Phi_i}{A_i} = J_i - H_i = J_i - \sum_{j=1}^N F_{ij} J_j$$

$$\Phi_i = \sum_{j=1}^N A_i F_{ij} (J_i - J_j) = A_i \frac{\epsilon_i}{1 - \epsilon_i} [E_{b,i} - J_i]$$

$$\Phi_i = \sum_{j=1}^N \frac{(J_i - J_j)}{\frac{1}{A_i F_{ij}}} = \frac{[E_{b,i} - J_i]}{\frac{1 - \epsilon_i}{A_i \epsilon_i}}$$



Electrical Analogy: Gray Surfaces

Small object in a large cavity

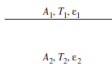


$$\frac{A_1}{A_2} = 0$$

$$F_{12} = 1$$

$$\dot{Q}_{12} = A_1 \sigma \epsilon_1 (T_1^4 - T_2^4) \quad (12-37)$$

Infinitely large parallel plates



$$A_1 = A_2 = A$$

$$F_{12} = 1$$

$$\dot{Q}_{12} = \frac{A \sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} \quad (12-38)$$

Infinitely long concentric cylinders



$$\frac{A_1}{A_2} = \frac{r_1}{r_2}$$

$$F_{12} = 1$$

$$\dot{Q}_{12} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1 - \epsilon_2}{\epsilon_2} \left(\frac{r_1}{r_2} \right)} \quad (12-39)$$

Concentric spheres



$$\frac{A_1}{A_2} = \left(\frac{r_1}{r_2} \right)^2$$

$$F_{12} = 1$$

$$\dot{Q}_{12} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1 - \epsilon_2}{\epsilon_2} \left(\frac{r_1}{r_2} \right)^2} \quad (12-40)$$